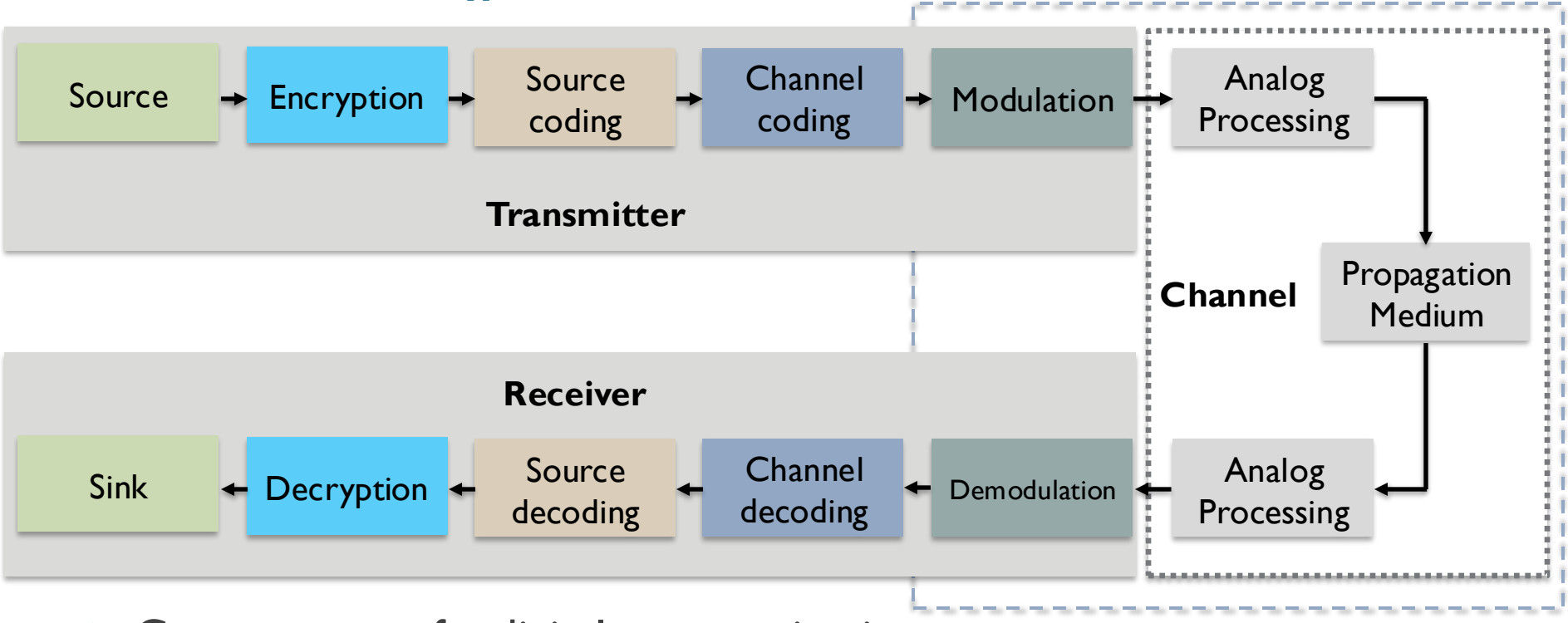
Probability and Random Variables

EE 471C / EE 381K-17 Wireless Communication Lab

From Lecture #2



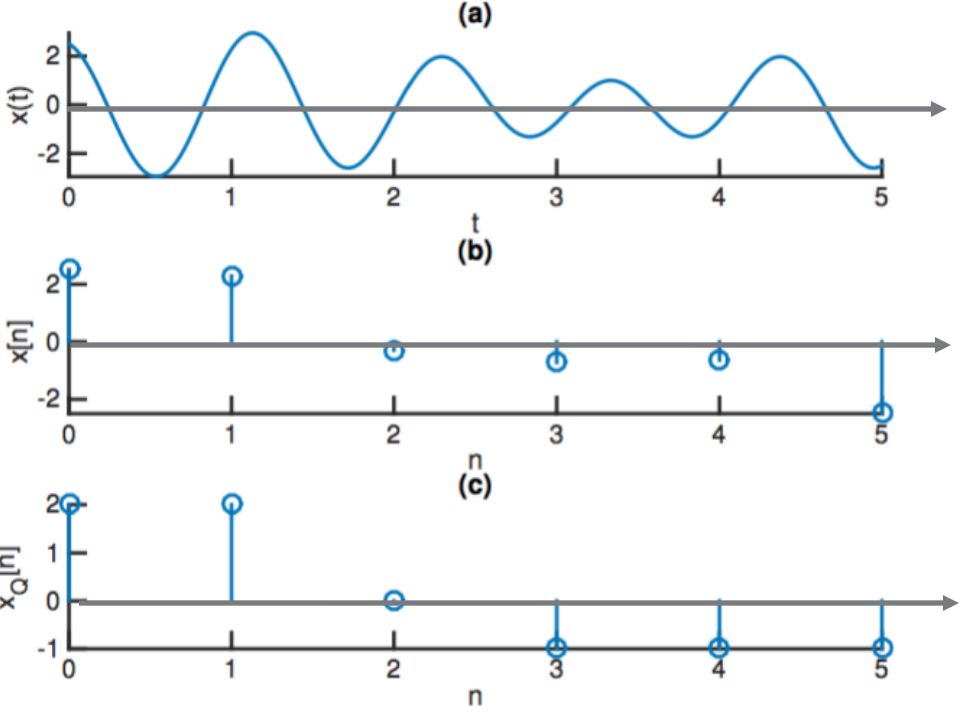
- Components of a digital communication system
- Lectures #3 - #6 mathematical background and models

Random signals

Learning objective:

- Explain why stochastic processes or random signals are useful in wireless communications.

Signals used in communication systems



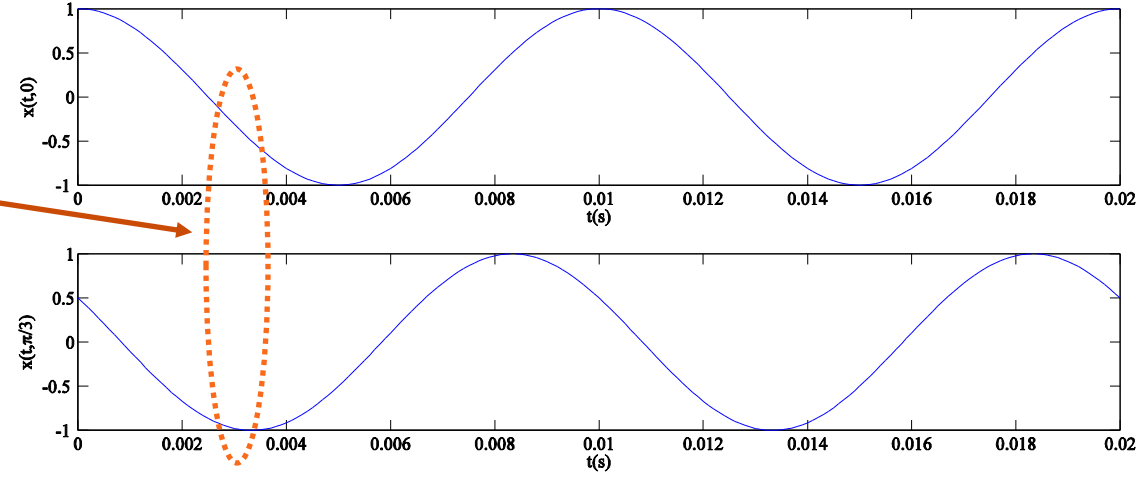
- Continuous-time x(t)
- Discrete-time x[n]
- Quantized $x_Q[n]$

Today: develop models for random signals

Random signals are modeled using stochastic processes

$$x(t,\theta) = cos(2\pi f_c t + \theta),\ \theta \sim U[0, 2\pi]$$

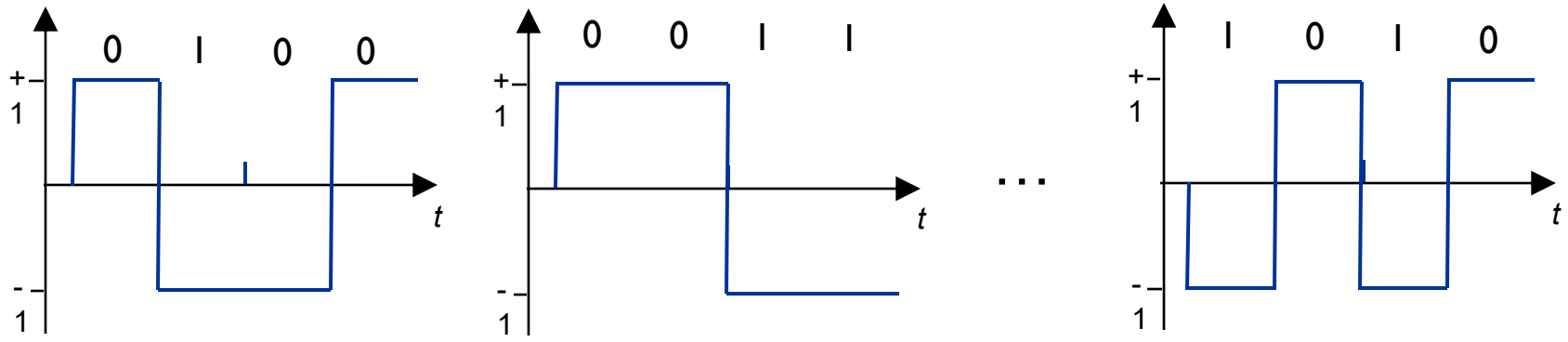
The phase at a given t is random



Two different realizations of the process

- Probabilistic model for a random signal, sequence, or function
- Characterizes an **ensemble** of random signals, not just a single realization
- Signal value at *any point of time* is a **random variable**

Why use stochastic processes?



- Used for receiver design, analysis of systems, and simulations
- Good model for the
  - Transmitted information
  - Thermal noise (most common receiver impairment known)
  - Propagation channel (changing environments over time)

Stochastic processes are built on probability

Random variables and probability

Learning objective:

- Compute the mean or variance of a continuous or discrete random variable

Continuous-valued random variables (RVs)

- Probability distribution: $f_x(z)$
- Cumulative distribution function: $F_x(\alpha) = \int_{-\infty}^{\alpha} f_x(z)\mathrm{d}z$
- Useful to compute the probability of a RV falling within a given range

$$P(x \leq \alpha) = F_x(\alpha)$$
$$P(x_1 \leq x \leq x_2) = F_x(x_2) - F_x(x_1) = \int_{x_1}^{x_2} f_x(x)dx$$

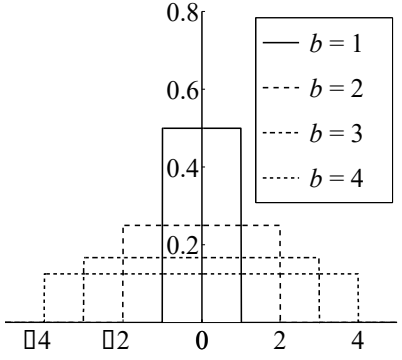
- Mean:

$$m_x = \mathbb{E}_x[x] = \int_{-\infty}^{\infty} z f_x(z)\mathrm{d}z$$

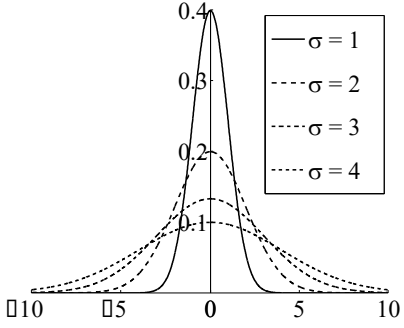
- Variance:

$$\sigma_x^2 = \mathbb{E}\left[|x - m_x|^2\right] = \mathbb{E}\left[|x|^2\right] - |m_x|^2$$

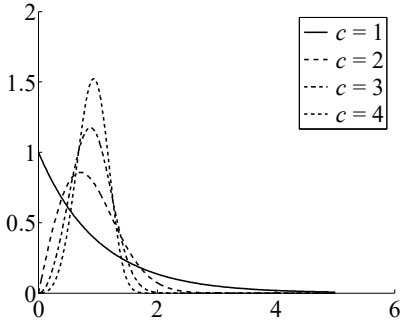
Example continuous distributions



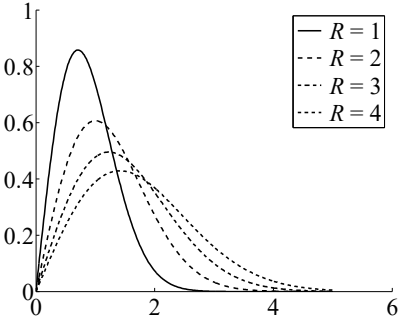
(a) Uniforme ($a = -b$)



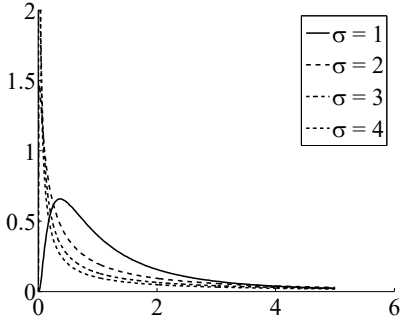
(b) Gausiana ($\mu = 0$)



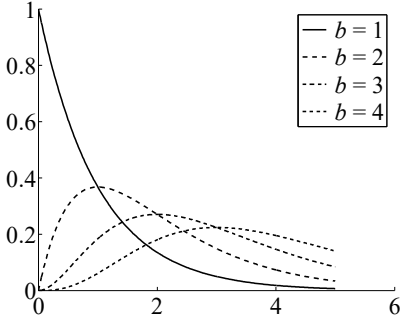
(c) Weibull ($\lambda = 1$)



(d) Rayleigh



(e) Log-normal ($\mu = 0$)



(f) Gamma

Discrete-valued random variables (RVs)

- Let $\mathcal{X}$ denote the set of discrete outcomes, indexed by m
- Let $p_x[m]$ denote the probability mass function of the m-th value
- $p_x[m]$ gives the probability that a discrete random variable x is exactly equal to some value $\mathcal{X}[m]$

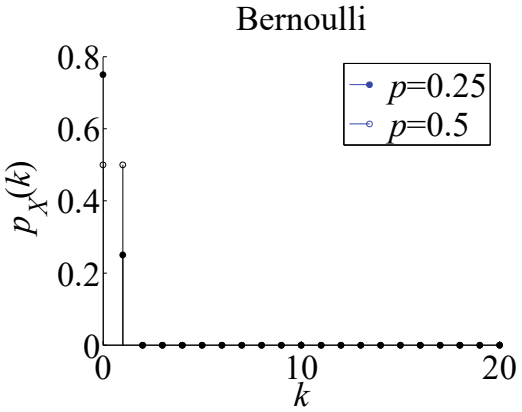
- Mean:

$$m_x = \sum_{m=-\infty}^{\infty} \mathcal{X}[m] p_x[m]$$

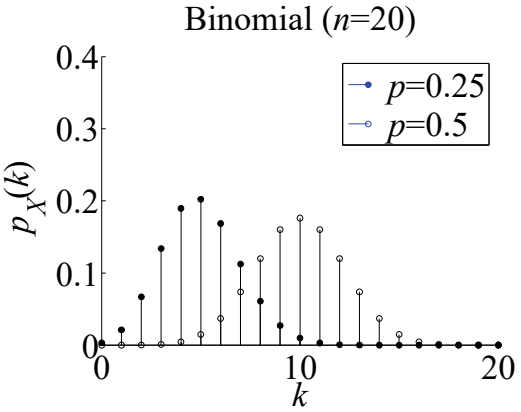
- Variance:

$$\begin{aligned} \sigma_x^2 &= \mathbb{E}\left[|x - m_x|^2\right] \\ &= \mathbb{E}\left[|x|^2\right] - |m_x|^2 \end{aligned}$$

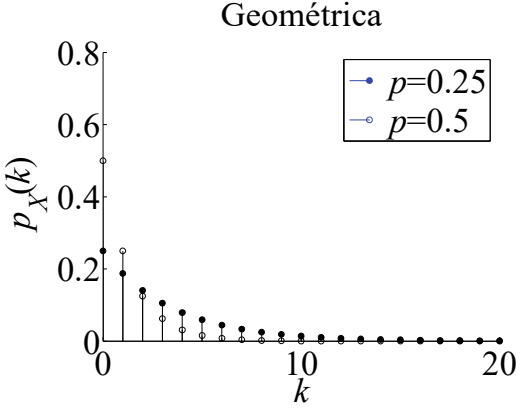
Example discrete distributions



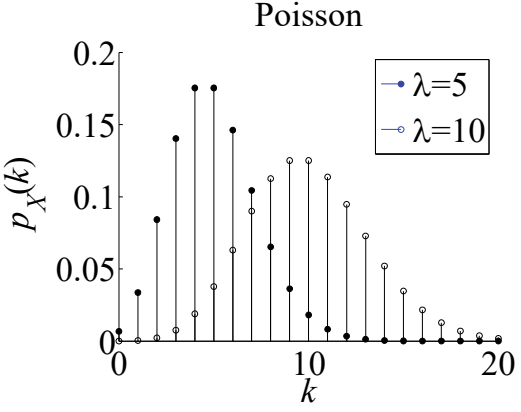
(a)



(b)



(c)



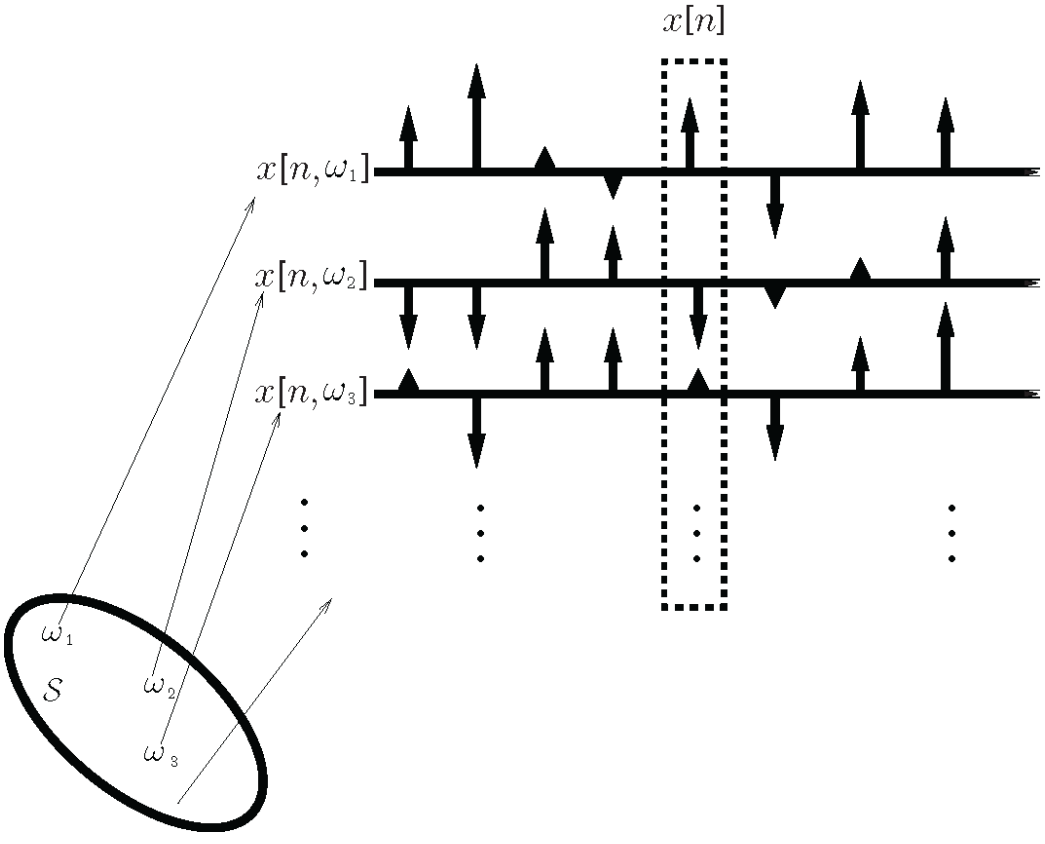
(d)

Stochastic Processes

Learning objective:

- Calculate mean, covariance, correlation of a stochastic process using expectation or ergodicity.
- Give some examples of stochastic processes in wireless.

Discrete-time random processes



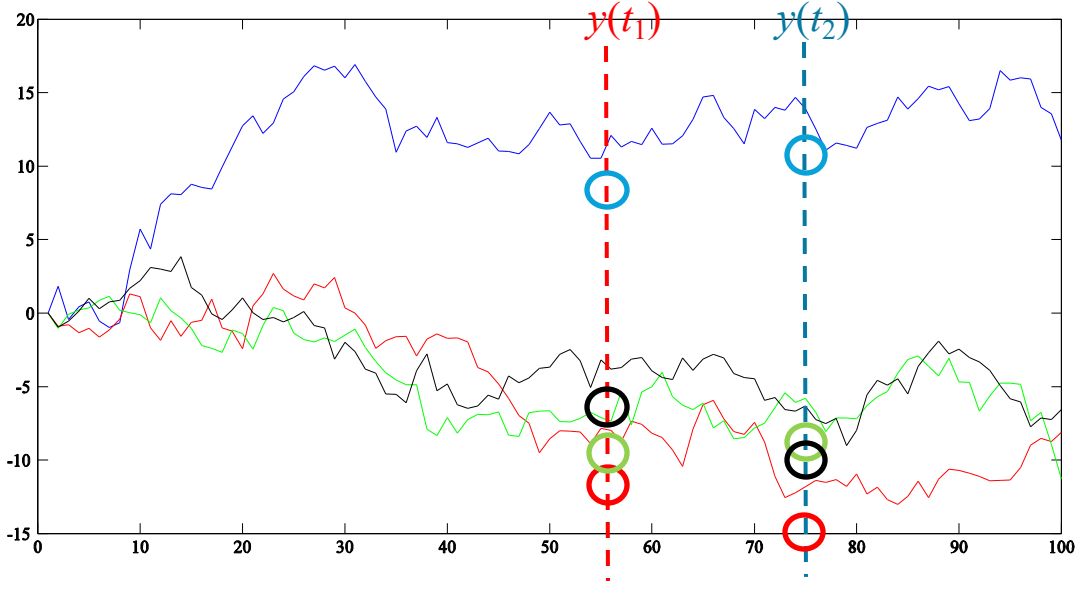
- A discrete-time random process (DTRP) is a sequence of random variables
- n is the time index
- x[n] is the random variable at n

Continuous-time random processes

Random walk

$$y(t) = y(t-1) + e(t),$$

$$e(t) \sim N(0,1)$$



- A continuous-time random process (CTRP) is a collection of random variables indexed by a real continuous number (can be generalized)
- Essentially, every x(t) is a random variable

Characterizations of a random process

- First order characterization of a DTRP is knowing the distribution for each n, i.e. $F_{x[n]}(\alpha)$
- Second order characterization of a DTRP is knowing the joint distribution between $x[n_1]$ and $x[n_2]$ for any choice of n_1 and n_2:

$$F_{x[n_1],x[n_2]}(\alpha_1, \alpha_2)$$

- Higher order characterizations can be similarly defined
- A random process is fully characterized by knowledge of all its higher order distributions … but this may not always be available

Independent identically distributed (IID) random process

- Most common kind of random process we will consider
- All the collections of random variables $(x[n_1], x[n_2], \ldots, x[n_k])$ for $k > 1$ are independent

$$F_{x[n_1],x[n_2],\ldots,x[n_m]}(\alpha_1, \alpha_2, \ldots, \alpha_m) = F_{x[n_1]}(\alpha_1) F_{x[n_2]}(\alpha_2) \cdots F_{x[n_m]}(\alpha_m)$$

- All the random variables are identically distributed thus

$$F_{x[n_k]}(\alpha_k) = F_x(\alpha_k)$$

for all $x[n_1], x[n_2], \ldots, x[n_k]$. Therefore

$$F_{x[n_1],x[n_2],\ldots,x[n_m]}(\alpha_1, \alpha_2, \ldots, \alpha_m) = F_x(\alpha_1) F_x(\alpha_2) \cdots F_x(\alpha_m).$$

Moments of a stochastic process (1)

- The mean of a random process is

$$\begin{aligned} m_x[n] &= \mathbb{E}_{x[n]}\left[x[n]\right] \\ &= \int_{-\infty}^{\infty} z f_{x[n]}(z) \mathrm{d}z \end{aligned}$$

- The correlation of a random process is

$$\begin{aligned} R_{xx}[n_1, n_2] &= \mathbb{E}_{x[n_1],x[n_2]}\left[x[n_1]x^*[n_2]\right] \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} z_1 z_2^* f_{x[n_1],x[n_2]}(z_1, z_2) \mathrm{d}z_1 \mathrm{d}z_2 \end{aligned}$$

where we use * to denote the complex conjugate

Moments of a stochastic process (2)

- The covariance of a random process is

$$C_{xx}[n_1, n_2] = \mathbb{E}_{x[n_1], x[n_2]} \left[x[n_1] x^*[n_2] \right] - m_{x[n_1]} m^*_{x[n_2]}$$

- If the process is zero-mean

$$C_{xx}[n_1, n_2] = R_{xx}[n_1, n_2]$$

- Most processes in the course will be zero-mean

Moments of a stochastic process (3)

- The variance (measure of the power of a random signal)

$$\sigma_x^2[n] = C_{xx}[n, n]$$

- A process if called uncorrelated* if

$$C_{xx}[n_1, n_2] = \begin{cases} \sigma_x^2[n_1], & n_1 = n_2 \\ 0, & n_1 \neq n_2. \end{cases}$$

- A process if called orthogonal if

$$R_{xx}[n_1, n_2] = \begin{cases} \mathbb{E}_{x[n_1]}|x[n_1]]|^2 & n_1 = n_2 \\ 0, & n_1 \neq n_2 \end{cases}$$

* Yes it seems strange this is not called uncovariated (if that is a word)

Cross correlation and covariance

- Cross correlation between two DTRPs is defined as

$$R_{xy}[n_1, n_2] = \mathbb{E}[x[n_1]y^*[n_2]]$$

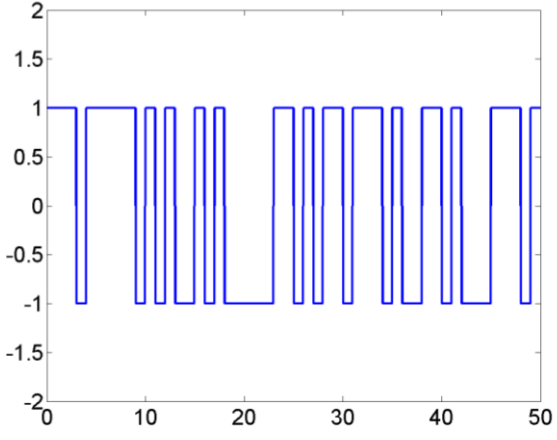
- Cross covariance between two DTRPs is defined as

$$C_{xy}[n_1, n_2] = \mathbb{E}[(x[n_1] - m_x[n_1])(y[n_2] - m_y[n_2])^*]$$

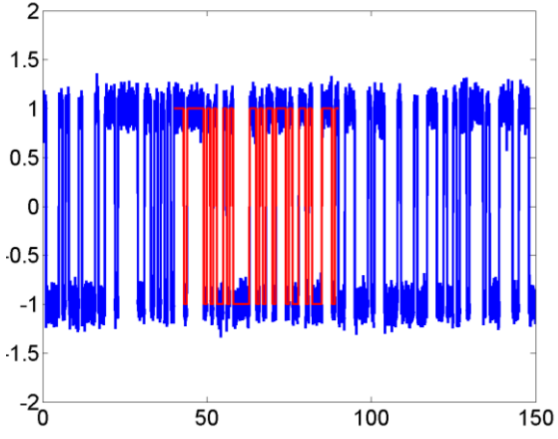
- These are measures of similarity between random processes

Cross correlation of a random processes

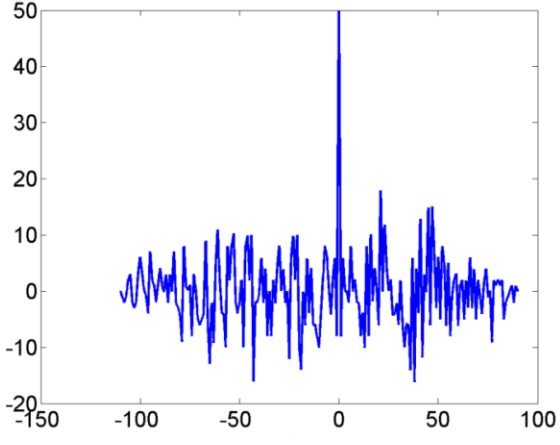
- In wireless communications cross correlation is used, for example, to:
  - Detect the header into the received noisy signal



Header



Received noisy signal



Cross correlation

  - Mitigate noise when detecting the transmitted sequence

Wide sense stationarity (WSS)

- A characterization of a random process based on its moments
- A wide sense stationary random process $\{x[n]\}$ satisfies:
  - The mean is constant: $m_{x[n]}[n] = m_x$
  - The covariance (or the correlation) is only a function of the difference:

$$C_{xx}[n_1, n_2] = C_{xx}[n_2 - n_1]$$

- No assumptions are made about joint probability distributions associated with a WSS random process
- By construction an IID random process is also WSS

IID Gaussian random processes (1)

- Most common model for thermal noise is the Additive White Gaussian Noise
- This refers to an IID random process where each sample is chosen from a Gaussian distribution with zero mean
- Real Gaussian distribution

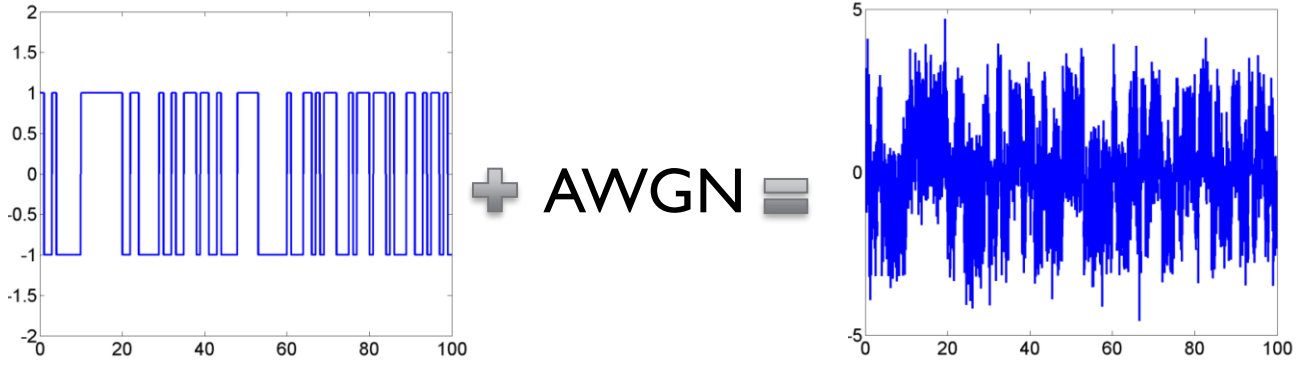
$$f_v(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{(x-m)^2}{2\sigma^2}}$$

  - Written in shorthand as $\mathcal{N}(m, \sigma^2)$
- The autocorrelation (and covariance) function of the AWGN is

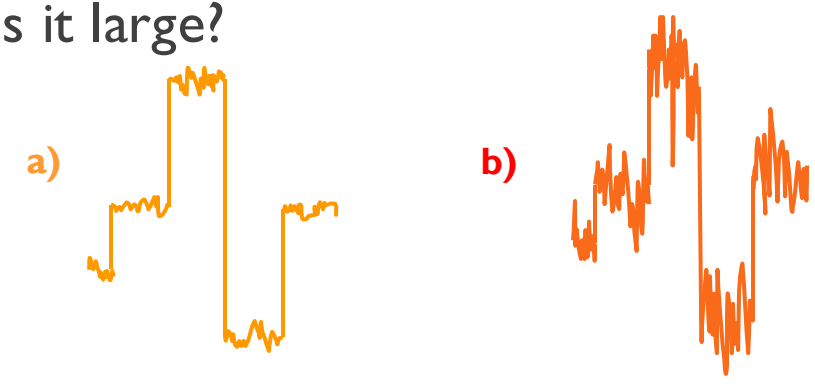
$$R_v[k] = \sigma^2 \delta[k]$$

IID Gaussian random processes (2)

- Example: digital modulated signal + white Gaussian noise



- If σ^2 is the variance of the AWG noise added to the modulated signal, in which case is it large?



IID Gaussian random processes (3)

- Also of interest circularly symmetric complex Gaussian distribution
  - Written in shorthand as $\mathcal{N}_{\mathbb{C}}(m, \sigma^2)$
  - Real and imaginary parts are independent with the same variance (but possible different mean) and distribution

$$f_v(x) = \frac{1}{\pi\sigma^2} e^{\frac{|x-m|^2}{\sigma^2}}$$

where x is complex

  - The relevant moments are

$$m_x = m$$
$$C_{xx}[n] = \sigma^2 \delta[n].$$

Filtering random processes

- Consider a WSS random process $x[n]$ and a LTI system
- At the output of the LTI system we have the random process

$$y[n] = \sum_{\ell=-\infty}^{\infty} h[\ell]x[n-\ell]$$

- It can be verified that the output is also WSS and

$$R_{xy}[k] = \mathbb{E}_x\left[x[n]y^*[n+k]\right] = h^*[k] * R_{xx}[k]$$

$$\mathbb{E}_y\left[y[n]\right] = \sum_{\ell=-\infty}^{\infty} h[\ell]\mathbb{E}_x[x[n-\ell]] \quad = m_x \mathsf{h}(e^{\mathrm{j}2\pi 0})$$

$$R_{yy}[k] = \mathbb{E}_x\{y[n]y^*[n+k]\} = h[-k] * h^*[k] * R_{xx}[k]$$

Ergodicity (1)

- One realization of the random process can be used to generate a reliable estimate of the moments
- With sufficient number of samples, sample averages converge to expectations

$$m_x = \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} x[n]$$

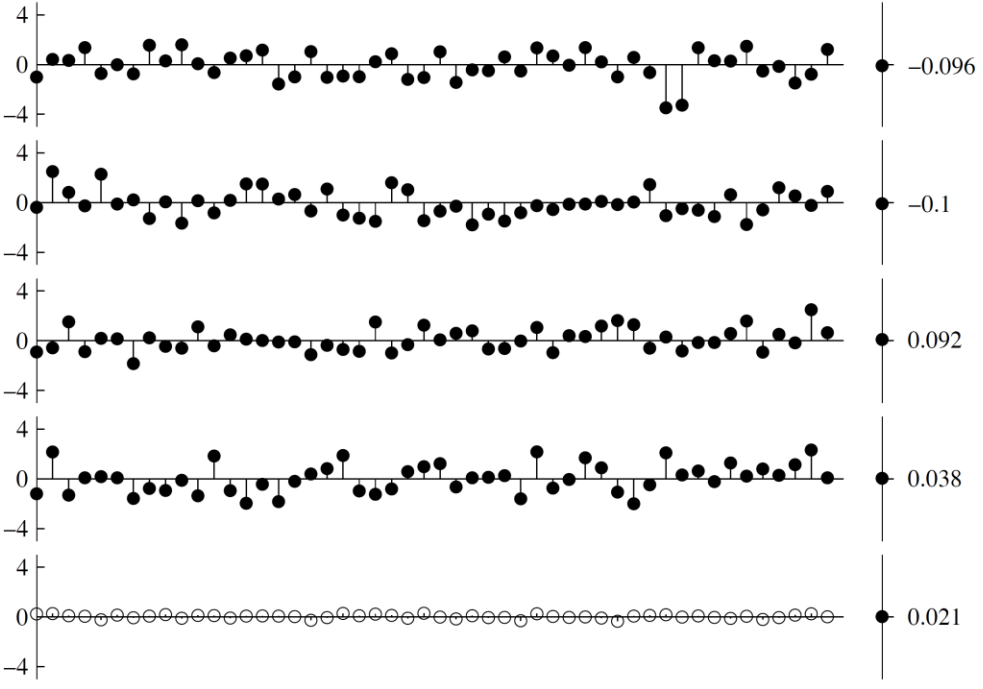
$$R_{xx}[k] = \lim_{N\to\infty} \frac{1}{N-k} \sum_{n=0}^{N-1-k} x[n]x^*[n+k]$$

- The covariance follows in a similar manner

- Most WSS RP are ergodic (exact conditions mentioned in the book)

Ergodicity (2)

- Consider four realizations of the random process in Exercise 2



- **Stationarity**
  average of each realization converges to mean
- **Ergodicity**
  average of four realizations converges to mean

Stochastic Processes

Learning objective:

- Calculate mean, covariance, correlation of a stochastic process using expectation or ergodicity.
- Give some examples of stochastic processes in wireless.